

Beam Polarimetry (for Future Experiments at JLab)

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¹JLab

PAVI-09

Outline

- 1 PV Program at 12 GeV - challenges
- 2 Electron Polarimetry
 - Compton Polarimetry
 - Møller Polarimetry
 - Møller with Atomic Hydrogen Target
- 3 Conclusion

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PV opportunities at 11-GeV

CEBAF is a excellent facility for PV experiments

PV at 6 GeV

- High polarization $\sim 85\%$
- High beam current $< 100\mu\text{A}$
- Low noise beam

Measured: G_s

Elastic $e p$, $e {}^4\text{He}$ (HAPPEX, G0)

Coming:

- Neutron skin ${}^{208}\text{Pb}$
 $e \text{Pb} \rightarrow e \text{Pb}$ (PREX)
- EW $e p \rightarrow e p$ (QWEAK)
- EW $e d$ DIS

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Higher energies:

$A \propto Q^2$ larger, but

σ_{elastic} suppressed by FF

Proposals:

- Møller PR-09-005 - app.
- DIS PR-09-012 - cond.app.

See talks by K.Kumar and P.Souder

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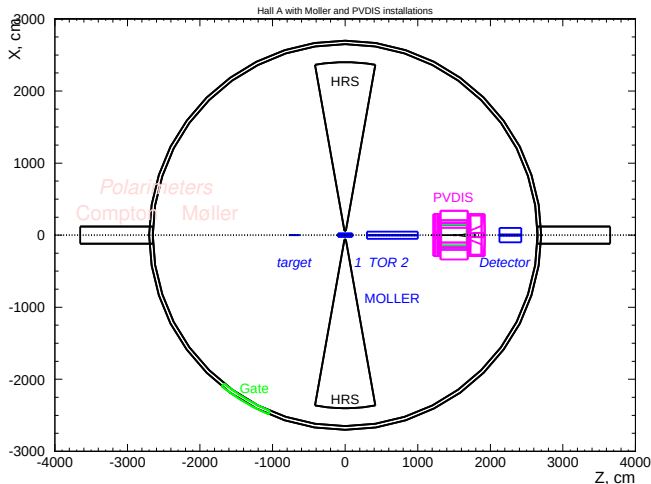
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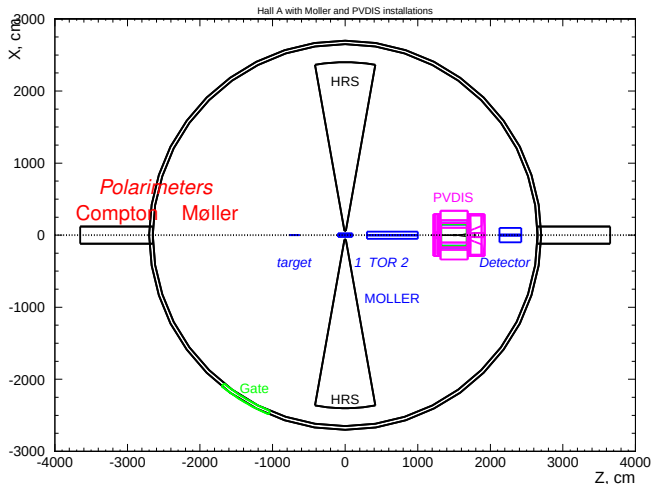
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Installation in Hall A



Installation in Hall A



Error Budget of Møller and PVDIS Experiments

Møller

Source of error	% error
Q^2 absolute value	0.5
beam polarization	0.4
beam second order	0.4
inelastic ep	0.4
elastic ep	0.3
other	0.5
total	1.0

PVDIS

Source of error	% error
beam polarization	0.4
radiative corrections	0.3
Q^2 absolute value	0.2
statistics	0.3
total	0.6

0.4% - can it be done?

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Electron Polarimetry for PV at JLab: Features

- Energy range $E_{beam} = 6.6 - 11 \text{ GeV}$
- Current range $I_{beam} = 40 - 90 \mu\text{A}$

Additional features to consider

- Time needed to achieve $\sim 0.4\%$ statistical error
- Systematic error:
 - Does polarimetry use the same beam as the experiment (energy, current, location)?
 - Continuous or intermittent (invasive?)
- Two different polarimeters/methods highly desirable

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Methods Used for Absolute Electron Polarimetry

Spin-dependent processes with a known analyzing power.

Atomic Absorption

$\vec{e}^- \sim 50 \text{ keV}$ decelerated to $\sim 13 \text{ eV}$ $\vec{e}^- + Ar \rightarrow Ar^* + e^-$, $Ar^* \rightarrow Ar + (h\nu)_\sigma$

Atomic levels: $(3p^5 4p)^3 D_3 \rightarrow (3p^6 4s)^3 P_2$ 811.5 nm fluorescence

Potential $\sigma_{syst} \sim 1\%$. Under development (Mainz) - only relative so far.

Currently - **invasive**, **diff. beam**

Spin-Orbital Interaction

Mott scattering, $0.1\text{-}10 \text{ MeV}$: $e^- \uparrow + Z \rightarrow e^- + Z$ $\sigma_{syst} \sim 3\%$, $\Rightarrow 1\%$ (?)

invasive, **diff. beam**

Spin-Spin Interaction

- Møller scattering: $\vec{e}^- + \vec{e}^- \rightarrow e^- + e^-$ at $>0.1 \text{ GeV}$,

$\sigma_{syst} \sim 1\text{-}2\%$, $\Rightarrow 0.5\%$

intermittent, mostly **invasive**, **diff. beam**

- Compton scattering: $\vec{e}^- + (h\nu)_\sigma \rightarrow e^- + \gamma$ at $>0.5 \text{ GeV} \sim 1\text{-}2\%$, $\Rightarrow 0.5\%$.

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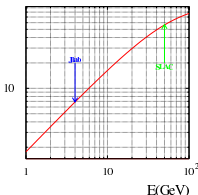
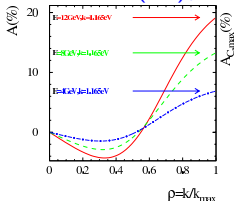
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Compton Polarimetry

$$\frac{\sigma_{\uparrow\uparrow} - \sigma_{\uparrow\downarrow}}{\sigma_{\uparrow\uparrow} + \sigma_{\uparrow\downarrow}} = A \cdot \mathcal{P}_b \mathcal{P}_t$$

Møller Polarimetry

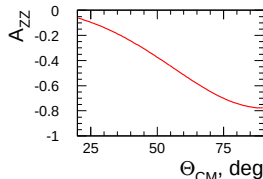
$$\vec{e}^- + (h\nu)_\sigma \rightarrow e^- + \gamma \text{ QED.}$$



- Rad. corrections to Born $< 0.1\%$
- Detecting: γ (0°), e^- $E < E_0$
- Strong $\frac{dA}{dk'}$ - good $\sigma E_\gamma / E_\gamma$ needed
- $A \propto kE$ at $E < 20$ GeV
- $T \propto 1/(\sigma \cdot A^2) \propto 1/k^2 \times 1/E^2$
- $\mathcal{P}_{laser} \sim 100\%$
- Non-invasive measurement

Syst. error 3→50 GeV: $\sim 1. \rightarrow 0.5\%$

$$\vec{e}^- + \vec{e}^- \rightarrow e^- + e^- \text{ QED.}$$



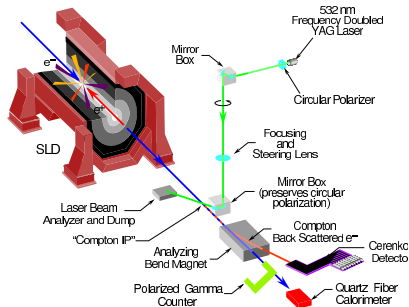
$$A(E) = -\frac{7}{9} \sigma_{lab} \sim 180 \frac{mb}{ster}$$

- Rad. corrections to Born $< 0.3\%$
- Detecting the e^- at $\theta_{CM} \sim 90^\circ$
- $\frac{dA}{d\theta_{CM}}|_{90^\circ} \sim 0$ - good systematics
- Beam energy independent
- Coincidence - no background
- Ferromagnetic target $\mathcal{P}_T \sim 8\%$
 - $\langle I_B \rangle < 3 \mu A$ (heating $1\%/100^\circ C$)
 - Levchuk effect (atomic e^-)
 - Low $\mathcal{P}_T \Rightarrow$ dead time
 - Syst. error $\sigma(\mathcal{P}_T) \sim 2\%$ (0.5%)

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SLAC SLD



source	$\sigma(P)/P$	
	SLD 1998	ILC Goal
Laser polarization	0.10%	0.10%
Analyzing power	0.40%	0.20%
Linearity	0.20%	0.10%
Electronic noise	0.20%	0.05%
total	0.50%	0.25%

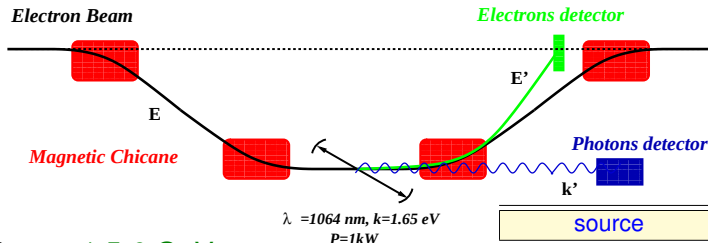
M.Woods, JLab Polarimetry workshop, 2003

- Beam: 45.6 GeV
- Beam: $3.5 \cdot 10^{10} e^- \times 120 \text{ Hz} \sim 0.7 \mu\text{A}$
- Laser: 532 nm, 50 mJ at 7 ns $\times$ 17 Hz
- Crossing angle 10 mrad
- e^- 17-30 GeV detector - gas Cherenkov
- γ detector - calorimeter
- Statistics 1% in 3 min

At JLab:

- Low energy: FOM: $\times 0.1$
- Coincidence e^-, γ
- Beam current $\times 100$

Compton Polarimeter in Hall A at JLab: CW cavity



- Beam: 1.5-6 GeV
- Beam: 5 – 100 μA at 500 MHz
- Laser: 1064 nm, 0.24 W
- Fabry-Pérot cavity $\times 4000 \Rightarrow 1 \text{ kW}$
- Crossing at 23 mrad $\varepsilon \sim 2.5\%$
- e^- detector - Silicon μ -strip
- γ detector - calorimeter

Stat: 1.0% 30 min, 4.5 GeV, 40 μA

Syst: 1.2% at 4.5 GeV

source	$\sigma(P)/P$
Laser polarization	0.50%
Response function	0.40%
Calibration	0.60%
Others	0.65%
total	1.15%

Upgrade - 1% at 1.0 GeV

- Laser: 532 nm, $\sim 1 \text{ W}$
- Cavity $\times 2000 \Rightarrow 2 \text{ kW}$
- Detector upgrade

Compton at Hall A: Expectations for 12 GeV

- Beam: 6.6-11 GeV
- Beam: 40 – 90 μA at 500 MHz
- Laser: 532 nm, 1 W
- Fabry-Pérot cavity $\times 2000 \Rightarrow 2 \text{ kW}$
- Crossing at 23 mrad $\varepsilon \sim 1.2\%$
- e^- detector - Silicon μ -strip
- γ detector - calorimeter
- High energy:
 - e^- far from the beam
 - Compton edge, $A=0$:
energy calibration

source	e^-	γ
Laser polarization	0.20%	
Analyzing power	0.20%	0.40%
Dead time	0.20%	0.00%
Pileup	0.00%	0.10%
Background	0.05%	0.05%
Others	0.03%	0.03%
Total	0.35%	0.46%

Stat: 1.0% 1 min, 11 GeV, 90 μA

Syst: 0.4% at 6.6 GeV

Seems hard but possible

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Møller Polarimetry

Polarized electron targets: magnetized ferromagnetic foils

- Iron: polarized d -shell (6 positions occupied out of 10)
- $\mathcal{P}_e$ not calculable: derived from measured magnetization
- Spin-orbital corrections ($\sim 5\%$) - measured in bulk material
- Magnetizing field is along the beam

Field 20 mT, foil at $\sim 20^\circ$

- Magnetization along the foil
- Magnetization can be measured
- A few % from saturation
- Sensitive to annealing, history
- Polarization accuracy $\sim 2 - 3\%$

Field 3 T, foil at $\sim 90^\circ$

- Magnetization perp. to the foil
- Magnetization - from world data
- Foil saturated
- Polarization is robust.
- Polarization accuracy $\sim 0.5\%$

Møller Polarimeter with Saturated Iron foil (Hall C)

JLab, Hall C, M. Hauger *et al.* NIM A **462**, 382 (2001), talk on PAVI09 by S.Page

- External $B_Z \sim 3 - 4 \text{ T}$

- Target foils $1\text{-}10 \mu\text{m}$, perp. to beam

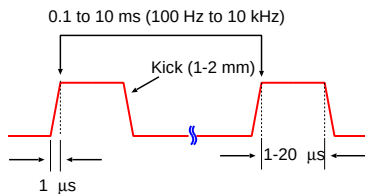
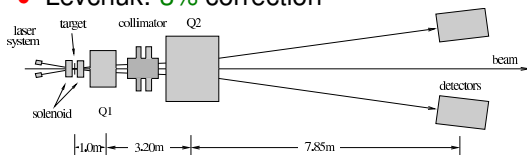
- $\mathcal{P}_t$ not measured

- Levchuk: 3% correction

Tests at high beam current

- Half-moon shape foil

- Kicker magnet

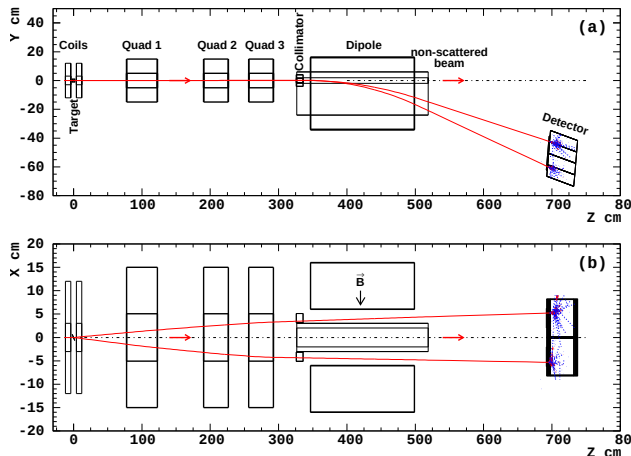


source	$\sigma(A)/A$
optics, geometry	0.20%
target	0.28%
Levchuk effect	0.30%
total at $3 \mu\text{A}$	0.46%
$\Rightarrow 100 \mu\text{A}$	?

A $1 \mu\text{m}$ thick half-foil: mech. problems:

- Foil unstable: holder design
- Thicker foil - high rate
- At $20 \mu\text{A}$ - accidentals/real ≈ 0.4

Hall A Møller Polarimeter



- Minimal Levchuk
- $\sigma_{stat} = 1\%$ in $\sim 2-3$ min
- $B_z \sim 25$ mT field
- Foil at 20° to field
- Foils $5-30\mu\text{m}$
- Beam $< 2\mu\text{A}$
- Systematics $\sim 2\%$

Hall A Møller Polarimeter: current upgrade

Upgrade, motivated by PREX requirements:

- High field magnetization (Hall C target clone)
- High instantaneous beam current:
reduce heating by introducing a beam duty cycle $< 5\%$
 - Beam repetition rate **500 MHz/4**
 - “Tune beam”: **4 ms** pulses at ~ 60 Hz
 - Instantaneous counting rate at **50 μ A** will be $\times 3$ higher
 - More invasive than a kicker scheme
- Electronics upgrade to digest higher rates

Møller Systematic Errors

Variable	Hall C	Hall A	
		Present	Upgrade
Target polarization	0.25%	2.00%	0.50%
Target angle	0.00%	0.50%	0.00%
Analyzing power	0.24%	0.30%	0.30%
Levchuk effect	0.30%	0.20%	0.20%
Target temperature	0.05%	0.00%	0.02%
Dead time	?	0.30%	0.30%
Background	?	0.30%	0.30%
Others	0.10%	0.30%	0.30%
Beam extapolation	?	larger	?
Total	0.47%	2.10%	~0.80%

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Possible Breakthrough in Accuracy

Møller polarimetry with 100% polarized atomic hydrogen gas, stored in a ultra-cold magnetic trap.

E.Chudakov and V.Luppov IEEE Trans. on Nucl. Sc., 51, 1533 (2004)

http://www.jlab.org/~gen/hyd/loi_3.pdf

Advantages:

- 100% electron polarization
 - very small error on polarization
 - sufficient rates $\sim \times 0.005$ - no dead time
 - false asymmetries reduced $\sim \times 0.1$
- Hydrogen gas target
 - no Levchuk effect
 - low single arm BG from rad. Mott ($\times 0.1$ of the BG from Fe)
 - high beam currents allowed: continuous measurement

Operation:

- density: $\sim 6 \cdot 10^{16}$ atoms/cm²
- Stat. error at 50 μ A: 1% in ~ 10 min

Møller Systematic Errors

Proposed: 100%-polarized atomic hydrogen target ($\sim 3 \cdot 10^{16}$ atoms/cm²).

Variable	Hall C	Hall A		
		Present	Upgrade	Proposed
Target polarization	0.25%	2.00%	0.50%	0.01%
Target angle	0.00%	0.50%	0.00%	0.00%
Analyzing power	0.24%	0.30%	0.30%	0.10%
Levchuk effect	0.30%	0.20%	0.20%	0.00%
Target temperature	0.05%	0.00%	0.02%	0.00%
Dead time	-	0.30%	0.30%	0.10%
Background	-	0.30%	0.30%	0.10%
Others	0.10%	0.30%	0.30%	0.30%?
Beam extapolation	?	larger	?	0.00%
Total	0.47%	2.10%	~0.80%	~0.35%

Hydrogen Atom in Magnetic Field

$H_1: \vec{\mu} \approx \vec{\mu}_e;$

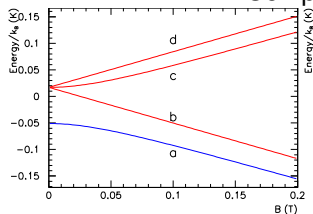
$H_2: \text{opposite electron spins}$

Consider H_1 in $B = 7 \text{ T}$ at $T = 300 \text{ mK}$

At thermodynamical equilibrium:

$$n_+/n_- = \exp(-2\mu B/kT) \approx 10^{-14}$$

Complication from hyperfine splitting:



Low energy

$$|b\rangle = |\downarrow\downarrow\rangle$$

$$|a\rangle = |\downarrow\downarrow\rangle \cdot \cos\theta - |\uparrow\downarrow\rangle \cdot \sin\theta$$

High energy

$$|d\rangle = |\uparrow\uparrow\rangle$$

$$|c\rangle = |\uparrow\downarrow\rangle \cdot \cos\theta + |\downarrow\uparrow\rangle \cdot \sin\theta$$

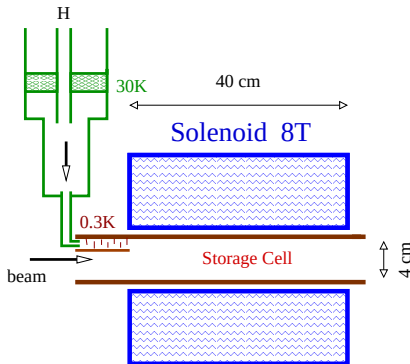
where $\tan 2\theta \approx 0.05/B(T)$, at 7 T $\sin\theta \approx 0.0035$

Mixture $\sim 53\%$ of $|a\rangle$ and $\sim 47\%$ of $|b\rangle$:

$$\mathcal{P}_e \sim 1 - \delta, \quad \delta \sim 10^{-5},$$

$$\mathcal{P}_p \sim -0.06 \text{ (recombination)} \Rightarrow \sim 80\%$$

Storage Cell



First: 1980 (I.Silvera,J.Walraven)
 $\vec{p}$ jet (Michigan)
 Never put in high power beam

- $-\vec{\nabla}(\vec{\mu}_H \vec{B})$ force in the field gradient
 - pulls $|a\rangle, |b\rangle$ into the strong field
 - repels $|c\rangle, |d\rangle$ out of the field
- $H+H \rightarrow H_2$ recombination (+4.5 eV)
 high rate at low T
 - parallel electron spins: suppressed
 - gas: 2-body kinematic suppression
 - gas: 3-body density suppression
 - surface: strong unless coated
 ~ 50 nm of superfluid ^4He
- Density $3 \cdot 10^{15} - 3 \cdot 10^{17} \text{ cm}^{-3}$.
- Gas lifetime > 1 h.

Contaminations and Depolarization of the Target Gas

Ideally, the trapped gas polarization is nearly 100% ($\sim 10^{-5}$ contamination).
 Good understanding of the gas properties (without beam).

Contamination and Depolarization

No Beam

- Hydrogen molecules $\sim 10^{-5}$
- Upper states $|c\rangle$ and $|d\rangle < 10^{-5}$
- Excited states $< 10^{-5}$
- Helium and residual gas $< 0.1\%$
 - measurable with the beam

100 μA Beam

- Depolarization by beam RF $< 2 \cdot 10^{-4}$
- Ion, electron contamination $< 10^{-5}$
- Excited states $< 10^{-5}$
- Ionization heating $< 10^{-10}$

Expected depolarization $< 2 \cdot 10^{-4}$

Gas Properties

- Atom velocity $\approx 80 \text{ m/s}$
- Atomic collisions $\approx 1.4 \cdot 10^5 \text{ s}^{-1}$
- Mean free path $\lambda \approx 0.6 \text{ mm}$
- Wall collision time $t_R \approx 2 \text{ ms}$
- Escape (10cm drift) $t_{es} \approx 1.4 \text{ s}$

CEBAF Beam

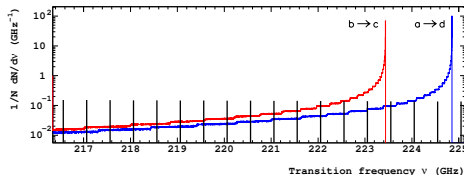
- Bunch length $\sigma = 0.5 \text{ ps}$
- Repetition rate 497 MHz
- Beam spot diameter $\sim 0.2 \text{ mm}$

Contaminations and Depolarization of the Target Gas

100 μA CEBAF beam:

Beam RF influence

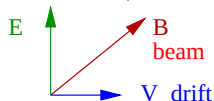
- $|a\rangle \rightarrow |d\rangle$ and $|b\rangle \rightarrow |c\rangle \sim 200 \text{ GHz}$
- RF spectrum: flat at $< 300 \text{ GHz}$



- $\sim 10^{-4} \text{ s}^{-1}$ conversions (all atoms)
- $\sim 6\% \text{ s}^{-1}$ conversions (beam area)
- Diffusion: contamination
 $\sim 1.5 \cdot 10^{-4}$ in the beam area
- Solenoid tune to avoid resonances

Gas Ionization

- 10^{-5} s^{-1} of all atoms
- $20\% \text{ s}^{-1}$ in the beam area
- Problems:
 - No transverse diffusion
 - Recombination suppressed
 - Contamination $\sim 40\%$ in beam
- Solution: electric field $\sim 1 \text{ V/cm}$
 - Drift $\vec{v} = \vec{E} \times \vec{B} / B^2 \sim 12 \text{ m/s}$
 - Cleaning time $\sim 20 \mu\text{s}$
 - Contamination $< 10^{-5}$
 - Ions, electrons: same direction
 - Beam $\vec{E}_r(160 \mu\text{m}) \approx 0.2 \text{ V/cm}$



Summary on Atomic Hydrogen for Møller Polarimetry

Potential for Polarimetry

- Systematic accuracy of $< 0.3\%$
- Continuous measurements
- Tools for systematic studies:
changing the electrical field (ionization)
changing the magnetic field (RF depolarization)

Problems and Questions

- Electrodes in the cell: R&D is needed
- Residual gas 0.1% accurate subtraction
Coordinate detectors: the interaction point?
- Atomic cross section (mean free path...) needs verification
- Cost and complexity

Conclusion

New PV experiments require a $\sim 0.4\%$ polarimetry.

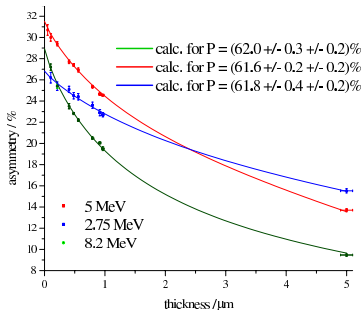
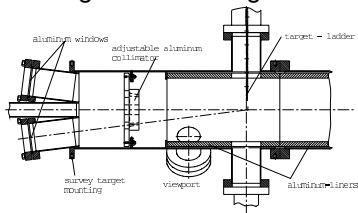
Possible strategy

- Continuous polarimetry
- $\sim 0.4\%$ Compton - seems feasible
- Second polarimetry method (Møller) of similar accuracy would help dramatically.
- Møller with iron targets - rather 1% than 0.4%
- Møller with atomic hydrogen target potentially can provide $< 0.30\%$ accuracy, very complex, needs R&D

Mott Polarimetry

0.1-10 MeV: $e^- \uparrow + Au \rightarrow e^- + Au$ analyzing power (Sherman func.) $\sim 1-3\%$

- Nucleus thickness: phase shifts of scat. amplitudes
- Spin rotation functions
- Electron screening, rad. corr.
- Multiple and plural scattering
- No energy loss should be allowed
- Single arm - background



- Extrapolation to zero target thickness
- $e^- \uparrow < 5 \mu A$ - extrapolation needed

JLab: $\sigma(P)/P = 1\%(\text{Sherman}) \oplus 0.5\%(\text{other})$ (unpublished) $\oplus \sigma(\text{extrapol})$

Dynamic Equilibrium and Proton Polarization

Proton polarization builds up, because of recombination of states with opposite electron spins:

$$|a\rangle = |\downarrow\uparrow\rangle\alpha + |\uparrow\downarrow\rangle\beta \text{ and}$$

$$|b\rangle = |\downarrow\downarrow\rangle$$

As a result, $|a\rangle$ dies out and only $|b\rangle = \downarrow\downarrow$ is left!

$$P \rightarrow 0.8$$

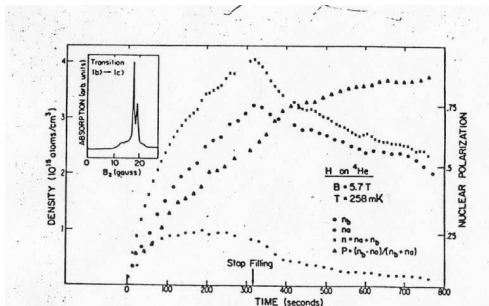


Fig. 5.11. Development of the densities of the hyperfine states and polarization as a function of time as measured by ESR. The inset shows the ESR line shape for one of the transitions (after van Yperen et al. 1983).